

ROMA TRE · A.Y. 2026/27

# Policy Evaluation Lab

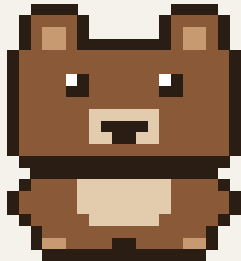
## *Lecture 3: Econometrics tutorial*

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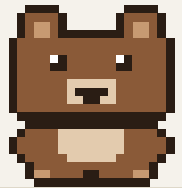
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PART 1

# Econometrics and regression

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# What is econometrics?

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**Econometrics** is the application of mathematical statistics to economic data to lend empirical support to the models constructed by mathematical economics and to obtain numerical estimates (Samuelson, 1954).

In short, the job of an applied economist is to:

- measure the relationship between variables (deduced or predicted by models)
- use such estimated relationships for predictions and forecasts



# What is a regression?

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- A **regression** allows you to estimate the impact of a variable of interest  $D$  on an outcome of interest  $Y$ , keeping other variables  $X$  constant
- Equation:

$$Y_i = \alpha + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + \gamma D_i + \epsilon_i$$

- $i$  denotes the unit of observation (e.g. country, region, firm, household, individual)
- $Y$  is called the **dependent variable** (or outcome of interest)
- $X$  and  $D$  are called the **independent variables** (or explanatory variables)
- $\epsilon$  is called the **error term**



## Adding the time dimension

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The time dimension may be added to the regression model to account for changes over time:

$$Y_{it} = \alpha_t + \beta_1 X_{1it} + \cdots + \beta_k X_{kit} + \gamma D_{it} + \epsilon_{it}$$

- $t$  might represent a year, quarter, month, day, etc.
- Observing the same unit  $i$  over time leads to **panel data**
- Different units sampled at different times result in **repeated cross-sections**



PART 2

# Data

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# Data and sample inference

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## Data collection:

- Ideally, you would ask everyone in the population of interest (e.g. the UK, the world), but this is often not possible
- Researchers typically work with a **sample** of observations from the population of interest
- Statistical methods are used to infer facts about the population from the sample

## The sample should be:

- a **random draw** from your population of interest, to ensure representativeness



## Data: cross-section

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| nr  | year | exper | married | educ |
|-----|------|-------|---------|------|
| 13  | 1986 | 7     | 0       | 14   |
| 17  | 1986 | 10    | 0       | 13   |
| 18  | 1986 | 10    | 1       | 12   |
| 45  | 1986 | 8     | 0       | 12   |
| 110 | 1986 | 11    | 1       | 12   |
| 120 | 1986 | 8     | 0       | 10   |
| 126 | 1986 | 8     | 1       | 13   |
| 150 | 1986 | 8     | 0       | 12   |
| 162 | 1986 | 8     | 0       | 11   |
| 166 | 1986 | 8     | 1       | 10   |
| 189 | 1986 | 8     | 0       | 14   |
| 193 | 1986 | 8     | 1       | 14   |
| 209 | 1986 | 9     | 0       | 11   |
| 212 | 1986 | 8     | 1       | 11   |
| 218 | 1986 | 8     | 1       | 15   |



## Data: panel

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| nr | year | agric | black | bus | construc | ent | exper | married | educ | tra |
|----|------|-------|-------|-----|----------|-----|-------|---------|------|-----|
| 13 | 1980 | 0     | 0     | 1   | 0        | 0   | 1     | 0       | 14   | 0   |
| 13 | 1981 | 0     | 0     | 0   | 0        | 0   | 2     | 0       | 14   | 0   |
| 13 | 1982 | 0     | 0     | 1   | 0        | 0   | 3     | 0       | 14   | 0   |
| 13 | 1983 | 0     | 0     | 1   | 0        | 0   | 4     | 0       | 14   | 0   |
| 13 | 1984 | 0     | 0     | 0   | 0        | 0   | 5     | 0       | 14   | 0   |
| 13 | 1985 | 0     | 0     | 1   | 0        | 0   | 6     | 0       | 14   | 0   |
| 13 | 1986 | 0     | 0     | 1   | 0        | 0   | 7     | 0       | 14   | 0   |
| 13 | 1987 | 0     | 0     | 1   | 0        | 0   | 8     | 0       | 14   | 0   |
| 17 | 1980 | 0     | 0     | 0   | 0        | 0   | 4     | 0       | 13   | 0   |
| 17 | 1981 | 0     | 0     | 0   | 0        | 0   | 5     | 0       | 13   | 0   |
| 17 | 1982 | 0     | 0     | 0   | 0        | 0   | 6     | 0       | 13   | 0   |
| 17 | 1983 | 0     | 0     | 0   | 0        | 0   | 7     | 0       | 13   | 0   |
| 17 | 1984 | 0     | 0     | 0   | 0        | 0   | 8     | 0       | 13   | 0   |
| 17 | 1985 | 0     | 0     | 0   | 1        | 0   | 9     | 0       | 13   | 0   |
| 17 | 1986 | 0     | 0     | 0   | 1        | 0   | 10    | 0       | 13   | 0   |



PART 3

# Ordinary least squares

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# Scatter plot and regression

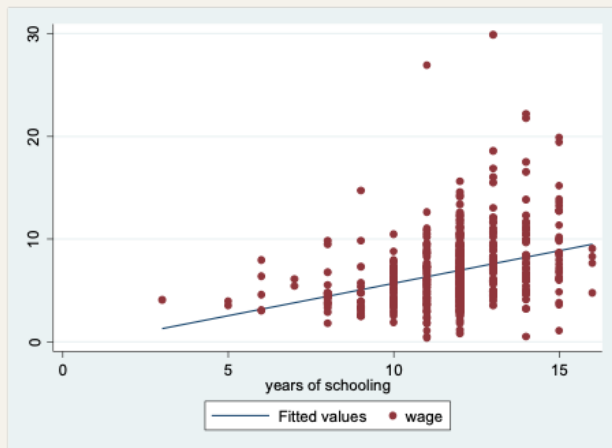
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- A **scatter plot** visualises the relationship between two variables, showing how one variable changes as the other changes
- In econometrics, the scatter plot helps us see the correlation between the independent and dependent variables
- Regression analysis, particularly **Ordinary Least Squares (OLS)**, is used to estimate the relationship between variables
- The OLS line minimises the sum of the squared vertical distances between the observed values and the regression line (the squared residuals)



# Scatter plot and OLS fitted line: wage and schooling

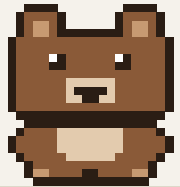
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PART 4

# Example: returns to education

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## Example and data

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**Example:** theory predicts that the wage ( $Y$ ) should increase with the level of education ( $D$ ).

**Empirical application:** estimating  $\gamma$  means knowing how the wage ( $Y$ ) of an individual  $i$  will change if they receive one additional year of education ( $D$ ), keeping all other variables ( $X$ , e.g. experience, ability, gender, marital status) constant.



# Results

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Results are usually presented in a table (produced with a statistical package such as R or Stata):

| Dep. Var.    | Wage (log)           |
|--------------|----------------------|
| Education    | 0.0878***<br>(0.012) |
| Time Dummies | Incl.                |
| N            | 545                  |
| R-sq         | 0.09                 |
| F-test       | 57.42***             |

Note: \*\*\* indicate statistical significance at the 1 percent level

- $N = 545$  is the size of the sample: we have 545 individuals
- $\hat{\gamma} = \partial \log(\text{wage}) / \partial D = 0.0878$



# Interpreting the results

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- The coefficient ( $\gamma$ ) indicates the effect of a one-unit change in the independent variable on the dependent variable
- **Standard errors** provide a measure of the precision of the coefficient estimate
- The  $R^2$  statistic measures how well the regression line fits the data
- Statistical significance is indicated with asterisks: \*\*\* (1% level), \*\* (5% level), \* (10% level)

A coefficient of 0.0878 with \*\*\* suggests that one additional year of education increases wages by approximately **8.8%** (the dependent variable is in logs), **statistically significant at the 1% level**



# Standard errors and significance (I)

| Dep. Var.    | Wage (log)           |
|--------------|----------------------|
| Education    | 0.0878***<br>(0.012) |
| Time Dummies | Incl.                |
| N            | 545                  |
| R-sq         | 0.09                 |
| F-test       | 57.42***             |

Note: \*\*\* indicate statistical significance at the 1 percent level

- The **standard error** (0.012, in brackets) gives you a sense of how precisely the relationship between  $D$  and  $Y$  is estimated (how much can we trust the estimate?)
- You want to know with which confidence you can say that the coefficient is different from zero, i.e. whether you can reject the **null hypothesis**  
 $H_0: \gamma = 0$
- If the sample is large enough (so that the estimator is approximately normal), when the coefficient is larger than its standard error by a factor of about 1.96 (in absolute value) we can reject  $H_0$  at the **5% significance level** (2.58 for the 1% level)



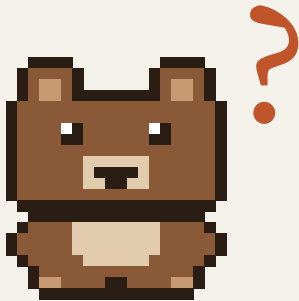
## Standard errors and significance (II)

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- 1.96 and 2.58 are critical values of the standard normal distribution, to which Student's  $t$  distribution converges in large samples
- \*\*\* tells you that we can reject the null hypothesis that the coefficient is equal to zero at the 1% significance level (if  $H_0$  were true, the probability of wrongly rejecting it would be at most 1%)
- \*\* and \* usually refer to the 5% and 10% significance levels (95% and 90% confidence)

\*\*\* means **statistically significant at the 1% level**, not a "99% chance" that  $\gamma \neq 0$





# Questions?

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